Dhruval Patel
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EDUCATION

Master of Computer and Information Sciences *(Aug 2024 - May 2026)*
The University of Texas at Dallas, Richardson, TX

Bachelor of Technology in Computer Science and Engineering *(Oct 2020 – Apr 2024)*
Indus University, Ahmedabad, India. ***(GPA: 9.84/10)***

PUBLICATIONS

- "Prediction of Heart Failure using Machine Learning Algorithms", Indian Journal of Natural Sciences, Tamil Nadu Scientific Research Organisation, **Web of Science Scopus Journal (WoS),** vol.14, issue 82. *(Feb 2024)*
- "A Comparative Review of Machine Learning Techniques for Predicting used Car Prices", International Journal of Research and Analytical Reviews, impact factor 7.17, [publication2]. *(Oct 2023)*
- "Advances and Challenges in Non-Invasive EEG-based Brain-Computer Interfaces: Applications, Technologies, and Future Prospects: A Comparative Review", International Journal of Advance Engineering and Research Development, [publication3]. *(May 2023)*

PROFESSIONAL EXPERIENCE

Indian Space Research Organisation, Space Application Centre, IN- Machine Learning Research Intern *(Jan-Jun 2024)*
- Engineered an advanced machine learning framework to reconstruct ocean chlorophyll concentrations at **10km pixel resolution** from satellite-simulated geospatial data reducing synthesis time by over 75%. Preserving seasonal and spatial variability with a minimal prediction error of 0.169 mg/m^3.
- Conducted comprehensive analysis on **600+ million flattened ROMS data**, utilizing advanced feature engineering, automated pipelines, and grid search to optimize predictive accuracy and accelerate execution. Applied **multi-layer correlation checks, spatial-temporal RMSE plots, time series visualizations,** and rigorous **sensitivity analysis** to extract granular insights into the model's feature adaptability.
- Collaborated with ISRO scientists, synthesized research outcomes into a peer-reviewed research paper, presenting findings through high-fidelity visualizations and technical reports, culminating in project completion within 50% of the allocated research timeline.

Maxgen Technologies, Ahmedabad – Data Science Intern *(Nov 2021 - Mar 2022)*
- Analysed real estate data encompassing **30+ attributes** across Sydney and Melbourne, developing predictive models to forecast property price trends, enabling clients to assess market dynamics and economic indicators for potential investments.
- Applied robust statistical diagnostics, including outlier detection, feature transformations and advanced visual analytics using Seaborn and Matplotlib to enhance data quality and interpretability.
- Constructed a systematic modelling framework integrating XGBoost, Support Vector Regression, Random Forest, and Polynomial Regression, applying iterative hyperparameter refinement to attain 93% predictive accuracy in price forecasting.

ACADEMIC PROJECTS

Cardiovascular Disease Prediction System
- Led development of ML models for heart failure prediction achieving **92% accuracy** using ensemble methods like Boosting Regression.
- Published research findings in Web of Science indexed journal and presented at BVM university conference which was ranked in top 5 projects.
- Developed **end-to-end Machine Learning pipeline** to process over **100,000 medical records**, integrating **Dask for parallel processing**, resulting in a **20% reduction in overall processing time**.

HyPrioriTEA: Intelligent Email Prioritization
- Engineered an automated pipeline leveraging Google Email API to process over 1,000 emails with custom prioritization rules.
- Adopted Spark and integrated MapReduce for parallel processing of text data, using TF-IDF and cosine similarity for robust content analysis.
- Deployed the solution on **Databricks Cloud**, achieving **40% faster processing time** through optimized **distributed computing**.

Story Generation from Multiple Images using NLP
- Developed ML pipeline integrating BLIP and Falcon models for image understanding and text generation.
- Built full-stack web application using Anvil framework with custom API endpoints.
- Implemented robust error handling and optimization for production deployment.

CERTIFICATIONS

- Diploma in Multi Lingual Computer Programming, Centre for Development of Advanced Computing *(Oct 2020 – Oct 2021)*
- Advanced PHP, C-DAC *(Dec 2021 - Feb 2022)*
- Online Courses: *IBM Data Analytics course, AWS Natural Language Processing, Google IT security: Defence Against Digital Dark Arts, Microsoft Presentation skills, Generative AI (Ollama, Llma, lava), AWS fundamentals, Databricks, Spark.*

PERSONAL ACHEIVEMENTS

Contribution in global Cancer Concern, Volunteering in mass awareness campaign and donation collection for cancer and aids, Caring Souls Foundation, Volunteering for Indian Student Association.

SKILLS

- Git, Python, SQL, Database Design, pandas, spark, map reduce, distributed computing, streamlit, ollama, Machine Learning, AWS fundamentals, Databricks, Geospatial Data Handling, Data visualization and analysis, Generative AI, Engineering Data, Big Data handling, predictive modelling, statistics, effective communication, team working spirit, bring up and deliver ideas, High enthusiasm, eagerness, credibility, High Approachability, Actively Social, highly passionate for social contribution, MS Powerpoint, Project Handling, management and version control, Agile, Scrum methodology.